

HYDROGENATED NBR (HNBR): PATENT LANDSCAPE ON SYNTHESIS, MOLECULAR-WEIGHT CONTROL, HYDROGENATION, AND POST-TREATMENT

1. ABSTRACT

This report synthesizes a 10-patent landscape focused on hydrogenated nitrile rubber (HNBR) derived from NBR precursors, with emphasis on hydrogenation chemistry, molecular-weight control, residual unsaturation, and downstream property optimization. The patents collectively define HNBR as hydrogenated acrylonitrile-butadiene rubber or related nitrile copolymers/terpolymers, typically based on butadiene and acrylonitrile, optionally with comonomers such as unsaturated carboxylic acids or esters. A recurring target is improved heat resistance and oil resistance while reducing Mooney viscosity and narrowing molecular-weight distribution without altering microstructure. Several patents address direct hydrogenation of NBR using homogeneous catalysts such as Wilkinson-type rhodium systems or ruthenium complexes, while others use heterogeneous catalysts such as silica-supported palladium. A second major theme is simultaneous metathesis/hydrogenation in the presence of hydrogen, using ruthenium/osmium catalysts to reduce molecular weight and polydispersity in a single step; these routes are contrasted with earlier two-step metathesis-then-hydrogenation processes and with co-olefin-assisted metathesis. Operating windows reported across the set include solution concentrations around 1-20 wt% polymer, temperatures from about 0 to 140 °C depending on route, and hydrogen pressures commonly around 3 MPa or 500-2000 psi in the hydrogenation examples. Catalyst loadings are generally low, often in the 0.01-0.3 wt% or phr range, with co-catalysts such as triphenylphosphine used in some systems. A distinct later-stage process patent focuses on removing residual palladium from hydrogenated nitrile rubber solutions by chelation with xanthates, dithiocarbamates, or trithiocarbonates followed by semi-coagulation and optional washing, enabling catalyst-free HNBR. Formulation patents then show how HNBR is compounded for high-temperature service using peroxide curing, unsaturated carboxylic acids/salts, polyamides, or microgels to improve hardness, rollability, compression set, and high-temperature tensile performance. Overall, the evidence supports HNBR synthesis strategies that balance high saturation, controlled residual double bonds, and molecular-weight retention with practical catalyst removal and compounding routes for durable elastomer applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10781272B2
- Patent Title: Process for recovery of residual hydrogenation catalyst from hydrogenated ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example EXAMPLE 1: - Example Extracted Example Example 6: - Example Extracted Example Example 5: - Example Extracted Example Example 4: - Example Extracted Example Example 3: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example EXAMPLE 8: - Example Extracted Example Example 14: - Example Extracted Example Example 13: - Example Extracted Example Example 12: - Example Extracted Example Example 11: - Example Extracted Example Example 10: - Example Extracted Example Example 9: - Example Extracted Example Example 8: - Example Extracted Example EXAMPLE 16: - Example Extracted Example EXAMPLE 18: - Example Extracted Example Example 19: - Example Extracted Example Example 20:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example example a: - Example Extracted Example Example 1::

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: US7381781B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2008
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 3: - Example Extracted Example Example 3:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US20100216961A1
- Patent Title: Hydrogenated nitrile rubber with narrow molecular weight distribution, a ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8: - Example Extracted Example Example 9:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: US7585920B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: EP2692788B1
- Patent Title: Highly saturated nitrile rubber composition and crosslinked rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative

Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Comparative Example 6: - Example Extracted Example Comparative Example 7: - Example Extracted Example Comparative Example 8: - Example Extracted Example Synthesis Example 1: - Example Extracted Example Synthesis Example 2: - Example Extracted Example Synthesis Example 3: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US5569719A
- Patent Title: Vulcanizable partially hydrogenated nitrile rubber composition containing ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1996
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example EXAMPLE 1: - Example Extracted Example EXAMPLE 2:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- The older route in US5569719A and US7381781B2/US7585920B2 uses partial hydrogenation of NBR to HNBR, typically with Pd/SiO₂ or Ru/Rh-type catalysts, targeting iodine values around 10-25 and residual double bonds below about 25%; these patents emphasize vulcanization performance or low-Mooney products rather than single-step molecular-weight reduction.
- US20100216961A1 and US9079979B2/US7585920B2/EP2473281B1/US7381781B2/US7951875B2 all center on simultaneous metathesis/hydrogenation or metathesis-first/hydrogenation-second concepts to lower Mw and Mooney viscosity; the key distinction is that US20100216961A1 and US9079979B2 explicitly perform metathesis and hydrogenation simultaneously in hydrogen, whereas EP2473281B1 and US7381781B2 describe metathesis followed by hydrogenation.

- Catalyst families differ strongly: US20100216961A1 and US9079979B2 rely on Ru/Os metathesis-capable complexes (Grubbs/Hoveyda/related), US7381781B2 and US7585920B2 use Ru/Os metathesis catalysts with hydrogenation capability, US7951875B2 uses hydrogenation catalysts including Hoveyda, Arlt, and Wilkinson's catalyst, and US5569719A uses Pd/SiO₂ for selective partial hydrogenation.
- Process conditions in the metathesis/hydrogenation patents are broadly similar: NBR cement in monochlorobenzene or similar inert solvents, catalyst loadings around 0.025-0.25 phr, hydrogen pressures around 500-2000 psi or 1200 psi, and temperatures from about 25 °C to 138 °C; US9079979B2 reports faster early hydrogenation with Grubbs III than Wilkinson's or Grubbs II at equal metal content.
- US10781272B2 is methodologically different from the synthesis patents: it does not make HNBR but instead removes residual palladium from already hydrogenated nitrile rubber solutions using chelating agents (PEX or SDEDTC), semi-coagulation with water, and optional washing; this is a downstream purification step rather than a polymerization/hydrogenation route.
- Formulation-focused patents EP2692788B1 and US9321910B2 move beyond synthesis to property tuning of highly saturated HNBR: EP2692788B1 combines two HNBR grades plus polyamide resin and peroxide curing, while US9321910B2 uses HNBR with unsaturated carboxylic acids/salts, peroxide, and microgels to improve high-temperature mechanical properties and demolding behavior.
- Historical progression is clear: early patents emphasize selective hydrogenation of NBR to HNBR and vulcanization performance; mid-generation patents introduce metathesis-assisted molecular-weight reduction; later patents refine catalyst choice, simultaneous hydrogenation/metathesis, and catalyst-removal/purification, followed by compounding strategies for high-temperature durability.

4. CONCLUSION

The patent evidence shows that hydrogenated NBR synthesis has evolved from straightforward selective hydrogenation of NBR to more sophisticated routes that simultaneously control saturation, molecular weight, and residual catalyst. The most robust synthesis strategies use NBR cements in inert solvents, low catalyst loadings, and hydrogenation conditions in the approximate range of 25-140 °C and 500-2000 psi, with Ru/Os metathesis-capable catalysts enabling single-step molecular-weight reduction and hydrogenation. For conventional HNBR production, Pd/SiO₂ or Rh-based hydrogenation remains effective, especially when the target is high saturation with retained molecular weight. For low-Mw HNBR, the strongest evidence supports simultaneous metathesis/hydrogenation in hydrogen, with NBR concentrations around 1-20 wt% and catalyst loadings around 0.025-0.25 phr. Post-hydrogenation purification by chelation/partial coagulation is a valuable enabling step when heterogeneous Pd catalysts are used. For downstream performance, peroxide curing with unsaturated carboxylic acids/salts, polyamide blending, or microgel incorporation improves high-temperature mechanical properties, rollability, and processability. Recommended synthesis parameters based on the evidence are: NBR

precursor with butadiene/acrylonitrile backbone, solution processing in monochlorobenzene or acetone/ketone media, low catalyst loading, hydrogen pressure near 1200 psi or 3 MPa for hydrogenation examples, and post-treatment to reduce residual palladium to near-zero when Pd catalysts are used.

5. REFERENCES

- US10781272B2 | Process for recovery of residual hydrogenation catalyst from hydrogenated ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10781272B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US7381781B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2008 | <https://patents.google.com/patent/US7381781B2/en>
- US20100216961A1 | Hydrogenated nitrile rubber with narrow molecular weight distribution, a ... | Not disclosed | US | 2010 | <https://patents.google.com/patent/US20100216961A1/en>
- US9079979B2 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9079979B2/en>
- US9321910B2 | Microgel-containing vulcanizable composition based on hydrogenated nitrile ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9321910B2/en>
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- EP2692788B1 | Highly saturated nitrile rubber composition and crosslinked rubber | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2692788B1/en>
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